

CS 566 COMPUTER VISION · FINAL PROJECT

How far off the recorded path can a Gaussian-splat driving simulator be trusted?

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PROBLEM

A splat is accurate only where the car actually drove



A Gaussian splat rebuilt from one lane, rendered 3 m to the side: **real photo (left) vs. render (right).**

Generative fixers (e.g., NVIDIA Difx) make such renders look realistic. **We measure whether they also make them correct.**

Real ground truth from three parallel lanes



Lane A: recorded, splat built here



Lane B (≈ 3 m): real photos exist



Lane C (≈ 6 m): real photos exist

Para-Lane scans the same roads in three parallel lanes. We build the splat from one lane and render it at the other lanes' recorded poses.

Trust = do a lane detector (CLRNet) and a vehicle detector (YOLO11) see the same thing in the render and in the real photo?

Hallucination = lane paint or vehicles in the render that the real photo does not have.

8 held-out test roads · 2,825 images at ≈ 3 m · 1,386 at ≈ 6 m · 95% bootstrap intervals.

What we compare, and what we built

Image fixers

- Released Difix (NVIDIA)
- NVIDIA Fixer (driving data)
- Difix training, re-implemented by us from the paper, trained on lane pairs
- Difix fine-tuned on our pairs
- Small U-Net from scratch

Fix the 3D scene

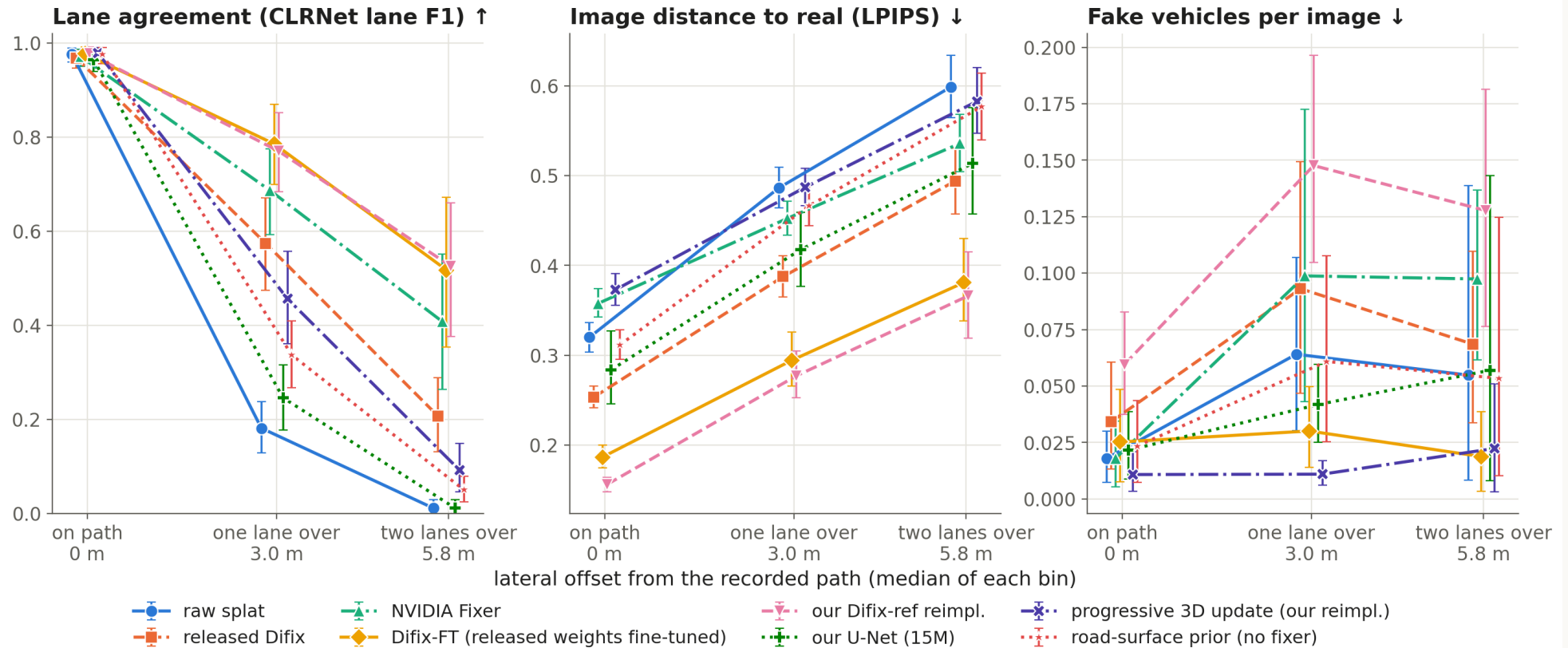
- Progressive update: fixed views added back into the splat (Difix3D+)
- Road-surface prior from LiDAR (ViSE, re-implemented; no generation)

Built and validated

- Our Difix trainer beats the official one at equal budget (DL3DV: LPIPS 0.218 vs 0.237)
- Our own splat projection: matches gsplat to 90.8 dB, gradients $\cos \geq 0.99996$

Trust collapses one lane over; trained fixers recover part of it

Trust vs. lateral offset, pooled over 8 held-out Para-Lane test roads (95% bootstrap CI over lane-pair sequences)



Lane agreement, raw render: 0.98 on path → 0.18 at 3 m → 0.01 at 6 m. **Best fixers at 3 m: 0.77–0.79.**

RESULT 2

Better-looking is not the same as correct

Hallucination, one lane over (4.1 m), Para-Lane scene_17

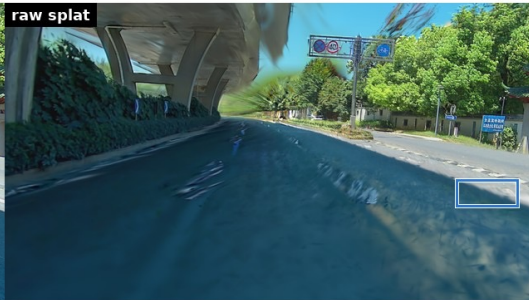
Rule (scene_16/17): among the 30 frames per fixer with the largest paint-precision drop vs. the raw splat, the largest invented marking (our Difix-ref reimpl.); marks use the metric's own paint masks.

real image at this pose



scene_17, lane 1 (real); splat trained on lane 2 only; offset 4.1 m

raw splat



lane F1 0.00 · paint P 0.95 / R 0.92 · LPIPS 0.423 · fake veh. 0

released Difix



lane F1 0.00 · paint P 0.57 / R 0.93 · LPIPS 0.412 · fake veh. 0

NVIDIA Fixer



lane F1 0.80 · paint P 1.00 / R 0.94 · LPIPS 0.378 · fake veh. 0

Difix-FT (fine-tuned)



lane F1 1.00 · paint P 1.00 / R 0.90 · LPIPS 0.314 · fake veh. 0

our Difix-ref reimpl.



lane F1 0.86 · paint P 0.13 / R 0.92 · LPIPS 0.271 · fake veh. 0

○ invented paint (not in the real image) □ erased paint (real marking missing) □ fake vehicle (YOLO11, no real object there)

Data: Para-Lane (Ni et al., 3DV 2025), CC BY-NC 4.0. CS 566 course project, non-commercial. Vehicles differ between passes (lanes driven at different times); metrics mask them out.

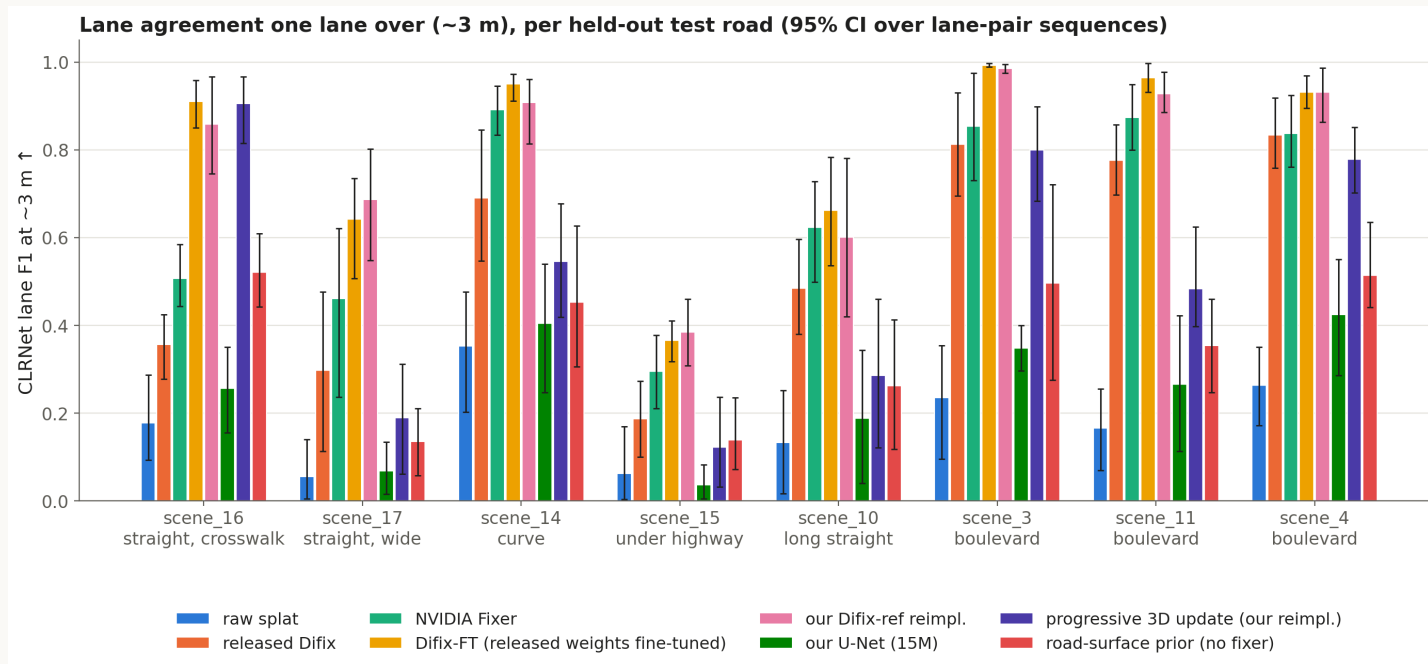
Our Difix paints a wide stripe that is not on the real road: paint precision 0.13 vs 0.95 raw, while its lane score stays high.

Fake vehicles: our re-implementation adds them on 8 of 8 roads (+0.09 per image); fine-tuning NVIDIA's weights does not.

Refining the 3D scene invents almost nothing (0.01 per image) but helps less far off the path.

RESULT 3

It depends on the road, and gains do not carry into driving



Per road: best fixer 0.95 on a curve, ≤ 0.39 under a highway; two lanes over there, every fixer draws a plausible but wrong road.

Camera height ± 1 m: -5 to -7 dB on the road; turning $\pm 15^\circ$ barely matters.

Other cities (EUVS, MV2): fixers trained on our roads do not transfer.

Closed loop (HUGSIM, 75 runs): no fixer improves the driving score (0.408 without; changes -0.05 to 0.00).

TAKEAWAYS

1. A splat-based simulator is trustworthy only near the recorded path: lane agreement falls from 0.98 to 0.18 one lane over.
2. Fixers trained on real neighboring-lane pairs recover most of it at 3 m (0.77–0.79), much less at 6 m (≈ 0.5).
3. Realism and correctness diverge: fixers invent lane paint and vehicles; repairing the 3D scene invents least.
4. Gains depend strongly on the road and do not transfer to other cities or to closed-loop driving scores.

Code, videos and all results: [project webpage](#).